Abdul Wahab

+1 647-836-5634 • wahab1@ualberta.ca
mirzaabdulwahab1612.github.io • LinkedIn

Education

- **University of Alberta** — **Edmonton, Alberta**
  MSc Computer Science supervised by Dr. Martha White — *2021 – 2023*
- **National University of Sciences and Technology** — **H-12 Islamabad**
  BSc Computer Science , Summa Cum Laude — *2016 – 2020*

Publications

K. Janjua, **A. Wahab**, B. Rashidi, **"Panoptic Pairwise Distortion Graph"**, in ICLR 2026. [Paper]

A. Wahab, R. Kumaraswamy, M. White, **"Value Bonuses using Ensemble Errors for Exploration in Reinforcement Learning,"** in Reinforcement Learning Conference 2025. [Paper]

A. Wahab, M. A. Tahir, N. Iqbal, A. Ul-Hasan, F. Shafait, S. M. Raza Kazmi, **"A Novel Technique for Short-Term Load Forecasting Using Sequential Models and Feature Engineering,"** in IEEE Access 2021 [Paper] [Code]

A. Ahmad, **A. Wahab**, F. Slyne, S. Zeb, R. Khan, M. Ruffini, **"Capacity sharing approaches in multi-tenant, multi-service PONs for low-latency fronthaul applications based on cooperative-DBA,"** in IEEE Optical Fiber Communications Conference and Exhibition (OFC) 2020. [Paper] [Poster]

Work Experience

- **AMII – Advanced Technology** — **Edmonton, Alberta**
  Machine Learning Scientist — *February 2026 – Present*
  I lead AI projects that apply reinforcement learning, large language models, agentic AI, and multi-agent systems to practical industry solutions. My work spans sequential decision-making, reasoning, information extraction, feedback incorporation, and AI agent evaluation across healthcare, smart grids, field service operations, and game environments.
- **Huawei Canada** — **Edmonton, Alberta**
  Associate Researcher — *February 2024 – February 2026*
  As an Associate Researcher, I worked on multiple components of the Image Signal Processing (ISP) pipeline for Huawei smartphone cameras, focusing on deep learning based solutions for image restoration, model optimization, and quality assessment. Below is a summary of the key projects I have contributed to:
  - **RL-Based Personal Assistant:** Lead the development of a simulation environment for diverse human routines and a lightweight RL agent that adapts online to user preferences for real-time recommendations on edge devices.
  - **LLM-Based Image Quality Assessment with Enhanced Vision Embeddings:** Improved existing image evaluation pipeline by adapting and extending VLMs, outperforming internal baselines by over 30%. Proposed a novel and easily pluggable graph-based distortion assessment method for pairwise images.
  - **Model Compression for Deployment-Efficient Inference:** Developed and delivered a structured pruning framework based on Dependency Graph , adapted to Huawei's proprietary architectures. Achieved a reduction from 130 GFLOPs to under 90 GFLOPs without performance loss.
  - **Domain Adaptation & Realistic Data Generation for RAW Denoising:** Addressed out-of-distribution generalization by developing a reference-based RGB-to-RAW conversion pipeline. Achieved 3–5% PSNR gains and substantial perceptual quality improvements on real test data. The toolkit is now adopted across image restoration teams.
  - **Image Denoising Architecture Development & Optimization:** Designed and implemented a deep learning–based image denoising module for Huawei smartphone cameras. Optimized the architecture by integrating frequency-based attention to reduce latency without compromising image restoration quality.

- **Stealth Health-Tech Startup** — **Remote**
  AI Research Engineer - Consultant — *October 2025 – December 2025*
  In a three-person team led by Prof. Matt Taylor, we delivered an MVP for an AI-assisted health-economics modeling pipeline. I developed the prompt-engineering and information-extraction components and contributed to simulation engine.

- **Veeve.io** — **Islamabad, Pakistan**
  Research Engineer — *July 2020 – July 2021*
  I developed machine learning and vision-based solutions for a smart shopping cart. I worked on cart state tracking, hand motion analysis, gesture prediction, shrinkage control, and bar-code detection and scanning through camera vision.

Research Experience

- **RLAI** — **Edmonton, Canada**
  Graduate Research Associate Fellow — *September 2022 - December 2023*
  - **Value Bonuses using Ensemble Errors for Exploration in Reinforcement Learning:** Developed a simple and scalable exploration algorithm that provides optimistic value bonuses for seen and unseen states.
  - **Understanding the Role of the Representations in Offline-Online Reinforcement Learning:** Studied the role of offline pre-training in the RL setting, and identified a simple enhancement: semi-gradient TD updates as an auxiliary task to help the agent learn better representations online.

- **Noah's Ark Lab, Huawei** — **Edmonton, Canada**
  Research Associate Intern — *May 2022 - August 2022*
  During this internship I started my research in the direction of sample-efficient exploration under the supervision of Dr. Raksha Kumaraswamy. We studied the role of on-policy and off-policy updates for uncertainty-based value bonuses.

- **Augmented Vision Lab (DFKI)** — **Kaiserslautern, Germany**
  Research Intern — *June 2019 - August 2019*
  I worked under the supervision of Prof. Dr. Didier Stricker and Dr. Gerd Reis , and developed a CNN-LSTM based architecture for line-scan cameras, for downstream tasks such as: road cracks segmentation, and seed sorting.

- **TUKL R&D Center** — **NUST SEECS, Islamabad**
  Research Assistant — *2018 - 2020*
  As a research assistant I contributed towards the following notable research projects:
  - **Line-scan camera feed processing with CNN-LSTMs (Thesis)**
  - **Short Term Energy Load Forecasting**
  - **Scheduling algorithm for multi-service, multi-tenant Passive Optical Networks**
  - **TCP's RTO prediction using sequence modeling for congestion control, and lower latency**

Academic Experience

- **Teaching Assistant**
  - **CMPUT-101 & CMPUT-174 Foundations of Computation** — **UoAlberta**
    Dr. Janelle Harms & Marianne Morris — *Fall'2021 & Winter'2022*
  - **CS-471 Machine Learning** — **NUST**
    Dr. Faisal Shafait — *Spring'2020*
  - **EE-353 Computer Networks** — **NUST**
    Dr. Arsalan Ahmad — *Fall'2019 & Spring'2019*
  - **CS-250 Data Structures & Algorithms** — **NUST**
    Dr. Faisal Shafait — *Fall'2018*

Skills

- **Programming:** C++, Python, Pytorch, Tensorflow, Scikit-learn, HuggingFace, OpenCV, Pillow
- **Frameworks:** Git, Linux